



NutriPurna: A Recipe Generator

By Shravani Halaye



Abstract

NutriPurna is a recipe generator that offers users a comprehensive culinary experience with features like nutrition tracking and meal planning tools. It provides refined search filters for recipes based on cuisine, meal type, cooking time, and calorie range. It includes a robust ingredient substitution tool to guide users in making healthy swaps. Additional features such as meal planning, nutrition tracking and a shopping list generator enhance meal preparation and dietary management.





Introduction

In today's fast-paced world, maintaining a healthy diet can be a challenge. The increasing demand for convenient and personalized meal solutions has led to the development of numerous recipe apps. However, many lack the ability to integrate with nutrient tracking apps, leading to a fragmented user experience. NutriPurna is introduced as a comprehensive solution an AI-powered recipe generator that not only assists users in discovering recipes tailored to their preferences and health needs but also integrates with tools for nutrient tracking and meal planning

1

Traditional Recipe generators

Traditional recipe generator offer a wide variety of recipes but often lack features and nutrient tracking capabilities.

2

Nutrient Tracking Apps

Nutrient tracking apps provide users with insights into their dietary intake but but don't help with actual meal decisions.

3

Integrated Solution

NutriPurna's Integrated Model offers both — combining recipe generation with in-depth dietary planning in one user-friendly platform.

Literature Survey

Sr. No.	Year	Author	Journal	Title of the paper	Result reported
1.	2023	Smriti Chaudhary,Binlta Soni, Aditya Sindhavad, Archi Mamaniya, Ashwini Dalwi, Irfan Siddavatam	IEEE XPLORE	ChefAI.IN : Generating Indian Recipes with AI Algorithm	The paper leverages AI and machine learning to create an autonomous recipe generator specifically for Indian cuisine, using a dataset of approximately 6,000 traditional recipes. The data was cleaned and preprocessed to ensure quality, and the AutoChef model which employs Autochef algorithm, was developed to generate new recipes based on user-selected ingredients. The primary algorithm used is a genetic algorithm, inspired by natural selection, which helps create unique and tailored recipes by selecting the best combinations of ingredients and cooking instructions.
2.	2020	Hajira Jabeen, Jonas Weinz, Jens Lehmann	IEEE XPLORE	AutoChef: Automated Generation of Cooking Recipes	The document evaluates AutoChef, which employs evolutionary algorithms and genetic programming to create valid and innovative recipes. It utilizes natural language processing to analyze existing recipes and generate new ones through a process involving data curation, ingredient detection, and recipe evolution. The system represents recipes as trees and explores new cooking methods by mutating these structures.
3.	2020	Yuran Pan, Qiangwen Xu, Yanjun Li	IEEE XPLORE	Food Recipe Alternation and Generation with Natural Language Processing Techniques	The research project investigates the use of natural language processing (NLP) techniques to assist in cooking by suggesting alternative ingredients and generating new recipes. It employs word embedding methods, specifically the Skip-gram model, to assess ingredient similarity, and utilizes language models such as the traditional N-gram and RNN with LSTM for recipe generation. The study analyzed a dataset of 3,433 recipes from various cuisines, revealing the potential of NLP to enhance cooking experiences.

Sr. No.	Year	Author	Journal	Title of the paper	Result reported
4.	2022	Khalid Azzimani, Hayat Bihri, Asma Dahmi, Salma Azzouzi, My El Hassan Charaf	IEEE XPLORE	An AI Based Approach for Personalized Nutrition and Food Menu Planning	Artificial intelligence (AI) and image recognition technology have been applied to develop systems for nutrient intake assessment and food recognition. These systems use classification, food detection, and segmentation processes to analyze images and identify food items. They aim to provide reliable data for planning therapeutic menus and assessing the effects of meals on patients under medical care. AI-powered applications like Calorie Mama and GoCARB have also been developed to accurately identify food and calculate calories for dietary management. The use of AI in personalized nutrition and precision medicine is gaining importance.
5.	2020	Trpit Sharma, Utkarsh Upadhyay, Ganesh Bagler	IEEE XPLORE	Classification of Cuisines from Sequentially Structured Recipes	The study evaluates various algorithms and models for cuisine classification based on recipes. It highlights that Logistic Regression achieved the highest accuracy of 57.70%, but models faced challenges due to a sparse dataset and high bias. Sequential models, particularly Transformers, outperformed traditional statistical models, with the RoBERTa model achieving the best accuracy of 73.3% and a loss of 0.10. The research also utilized LSTMs and emphasized the importance of considering cooking processes and techniques in addition to ingredients for improved classification accuracy.
6.	2016	Xuehui Mao, Shzhong Yuan, Weimin Xu, Daming Wei	IEEE XPLORE	Recipe Recommendation Considering the Flavour of regional cuisine	The document proposes a scoring method using cosine theorem for regional recipe recommendations that considers both the preferences of regional cuisines and the similarities between them. It quantify ingredient preferences of a regional cuisines by TF-IDF (Term Frequency - Inverse Document Frequency) and then score the dishes. It compose the score of the regional recipes and flavour similarity between regional cuisines into a recommendation using the Tidal-Trust algorithm

Existing System

Current recipe generators and nutrition tracking platforms typically offer basic features such as ingredient or cuisine-based search, nutritional information, and simple meal planning. However, they often lack advanced capabilities like dynamic nutrient tracking, intelligent ingredient substitution, and integration with personalized health data. Many rely on outdated databases and provide generic recommendations, with limited support for allergies, chronic conditions, or culturally relevant recipes. These gaps hinder their effectiveness in delivering a fully tailored and sustainable dietary experience.

Problem Statement and Objectives



Problem Statement

There is demand for platforms offering personalized recipes, nutrient tracking, and meal planning is increasing, yet many existing systems fail to provide customized solutions for a wide range of dietary preferences and health conditions. Users often face challenges in managing daily nutrient intake, finding appropriate recipes, and planning meals effectively, especially when navigating specific health concerns like diabetes or food allergies.

Objectives

- **Recipe Generation:** To design an intelligent system that allows users to generate personalized recipes based on available ingredients, dietary preferences, & health conditions.
- **Nutritional Analysis:** To integrate algorithms capable of accurately calculating macronutrient and micronutrient values from recipes and helping users maintain balanced diets.
- **Meal Planning:** To assist users in health-conscious meals through a dynamic and user-friendly interface that not only generates personalized recipes but also compiles a comprehensive 7-day weekly meal plan.

Proposed System

The proposed recipe generator utilizes a combination of data from NutriPurna, user preferences, and techniques to generate nutritionally balanced recipes.

1

User Input

Users begin by entering personal preferences such as diet type (vegetarian, diabetic, gluten-free), health conditions, ingredient availability, and nutrition goals.

2

Recipe Generation

An LLM analyzes the input and generates detailed recipes, complete with steps, cooking time, ingredients, and substitutes. It ensures variety and nutritional balance by considering cultural relevance and dietary guidelines.

3

Nutrient Tracking

Each recipe is evaluated using a nutrient calculation algorithm powered by third-party APIs (like Edamam). The nutritional values (calories, carbs, protein, fat, etc.) are stored and visualized on a tracker, allowing users to monitor their daily intake and progress toward their goals.



System Architecture

The system architecture consists of a front-end user interface, a backend server, and an integrated database that stores user data, recipe information, and nutrient profiles.



User Interface

A user-friendly interface allows users to browse recipes, personalize their preferences, and manage their dietary goals.



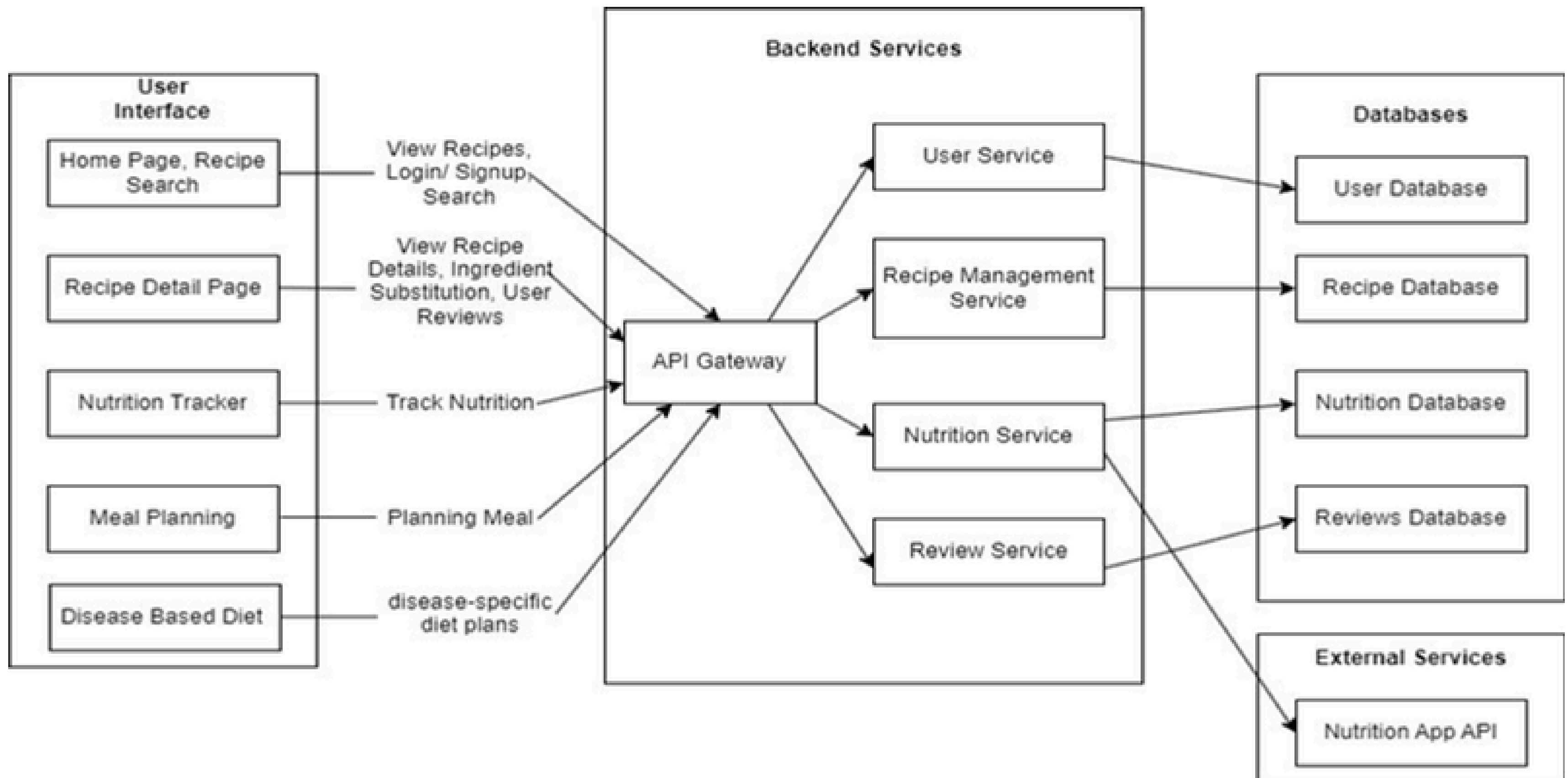
Backend Server

The backend server handles data processing, recipe generation, and communication with API.



Database

The database stores user data, recipes, and nutritional information, facilitating data access and retrieval.



System Architecture

Methodology :



1

Requirement Analysis

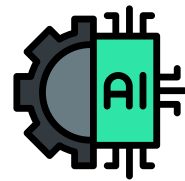
Research user needs, expectations, and market gaps for feature development.



2

System Design

Develop MVC architecture and database schema with tables for users, recipes, nutrients, and health guidelines.



3

Algorithm Design

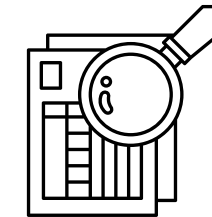
Create algorithms for recipe generation, nutrient tracking, meal planning, and health-based dietary recommendations.



4

Implementation

Build the frontend using HTML/CSS/JavaScript and backend using Python/Flask with database and API integration.



5

Testing and Validation

Perform device testing, gather user feedback, and refine usability.



6

Deployment & Maintenance

Launch the platform with optimizations for performance & accessibility. Provide regular updates, user support, & troubleshoot issues.

Algorithm :

LLM(Large language model)

NutriPurna utilizes a Large Language Model (LLM) to deliver highly personalized recipe recommendations based on user preferences, dietary needs, and ingredient availability. Unlike traditional filtering methods, LLMs interpret natural language inputs dynamically, ensuring flexibility and precision.

How It Works:

1. User Input Processing:
2. Recipe Matching Using LLM:
3. Generating the Recipe Output:



Nutrient Calculation Algorithm

In NutriPurna calculates the nutritional content of a recipe by retrieving ingredient data, adjusting it based on quantities, and summing the total values. It then adjusts the content per serving and compares it to Recommended Daily Values (RDVs), helping users track nutrient intake and meet their dietary goals.



Algorithm :

Rule-Based Approach

It relies on predefined rules that dictate how a system behaves using if-then logic. The rules are created to handle specific conditions, and the system uses an inference engine to evaluate which rule applies based on user inputs. Once a rule is matched, it executes the corresponding action, such as recommending recipes that match user preferences.



NutriPurna's Rule-Based Algorithm offers:

1. Disease-Based Meal Planning – Select a disease to get dietary recommendations and a 1-2 day meal plan.
2. Symptom-Based Search – Enter symptoms to identify diseases and receive dietary guidance.
3. Personalized 7-Day Meal Planning – Get a tailored weekly meal plan based on health profile and dietary needs.



System Requirements

Hardware Requirements:

1. RAM : 4GB minimum or more
2. Processor : Intel i3 minimum or more

Software Requirements:

- 1.OS – Windows 10, 11
- 2.Html, CSS, Javascript,Python
- 3.Server
- 4.Database – Firebase,SQLite
- 5.API – Google, Spoonacular
- 6.VS Code



Implementation Details for Recipe Generator

- **Frontend**

The frontend which contain features like dynamic search filters, recipe cards, feature images, nutritional information, save/bookmark options and a responsive user interface, ensuring seamless navigation and an engaging user experience.

- **Backend**

The backend will contain robust API endpoints for recipe search generation, and ingredient substitution. It will also manage user authentication and integrate with third-party nutrition databases to offer nutritional analysis.

- **Database**

The application utilizes structured databases to manage and store data such as user profiles, recipes, dietary restrictions, nutrition records, and substitution rules.



Implementation Details for Recipe Generator

- **APIs**

NutriPurna uses an API gateway to efficiently manage requests and enable communication between modules. APIs are integrated for real-time nutritional analysis, recipe data parsing, and authentication.

- **Additional Features**

1. Nutrition tracker to monitor daily intake and track health progress.
2. 7-day meal planner to organize balanced meals throughout the week.
3. Disease-based meal suggestions tailored to specific health conditions.
4. AI-powered chatbot
5. healthy ingredient alternatives and real-time diet tips.



Key Features

Recipe Generation from Ingredients

AI-generated recipes based on available ingredients and user preferences.



Nutritional Insights + Tracker

Track calories and nutrients with detailed breakdowns for every recipe.

Meal Planning with Diet Goals

Plan personalized weekly meals aligned with health and fitness goals

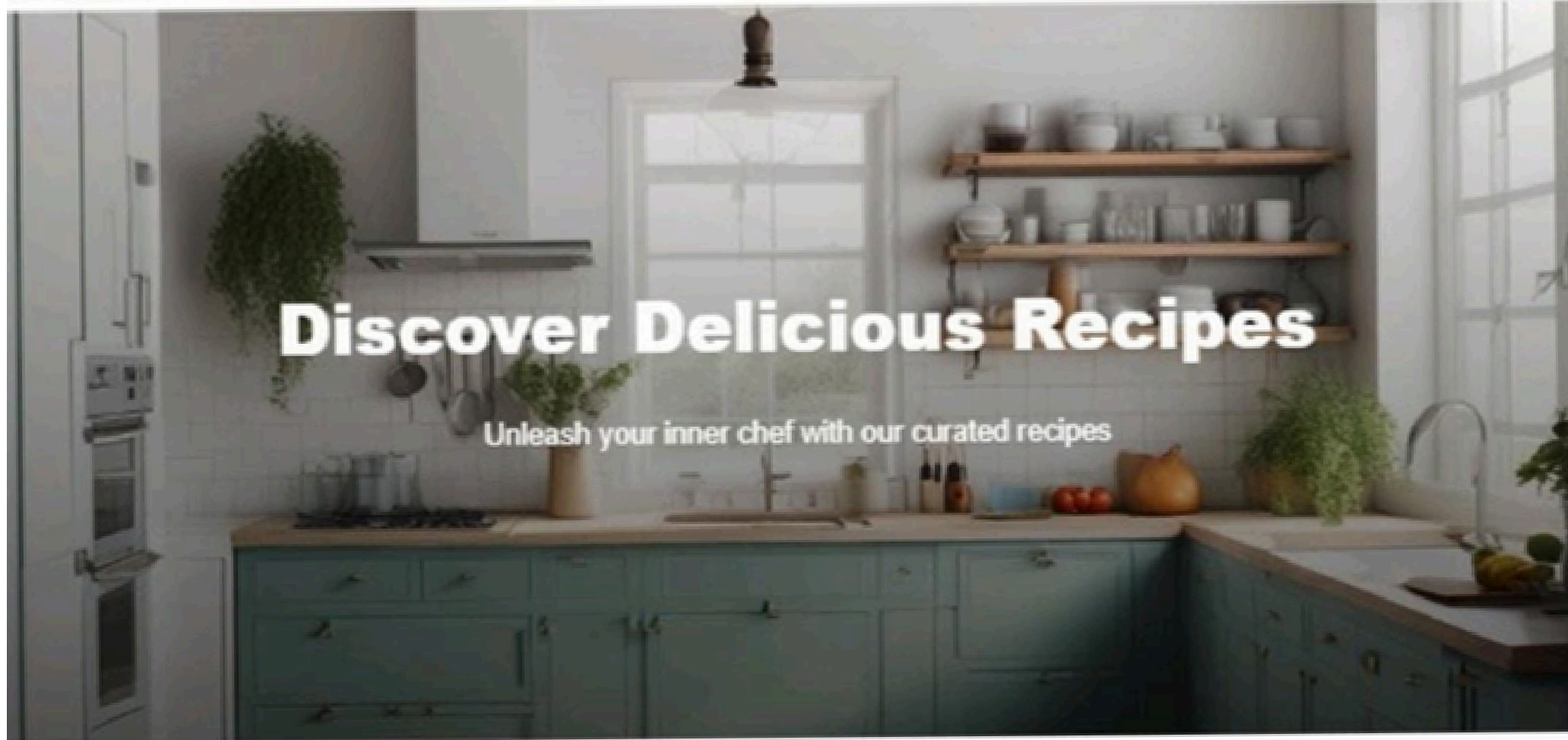


Disease-Based Diet Planner + Nutrition Chatbot

Get diet recommendations tailored to medical conditions like diabetes.

Screenshots

(UI Pages)



Cuisine

Any

Meal Type

Any

Diet

Any

Preparation Time

10 mins

20 mins

30 mins

45 mins

60 mins

90 mins

Extra

Serving Counts

1

2

4

8

12

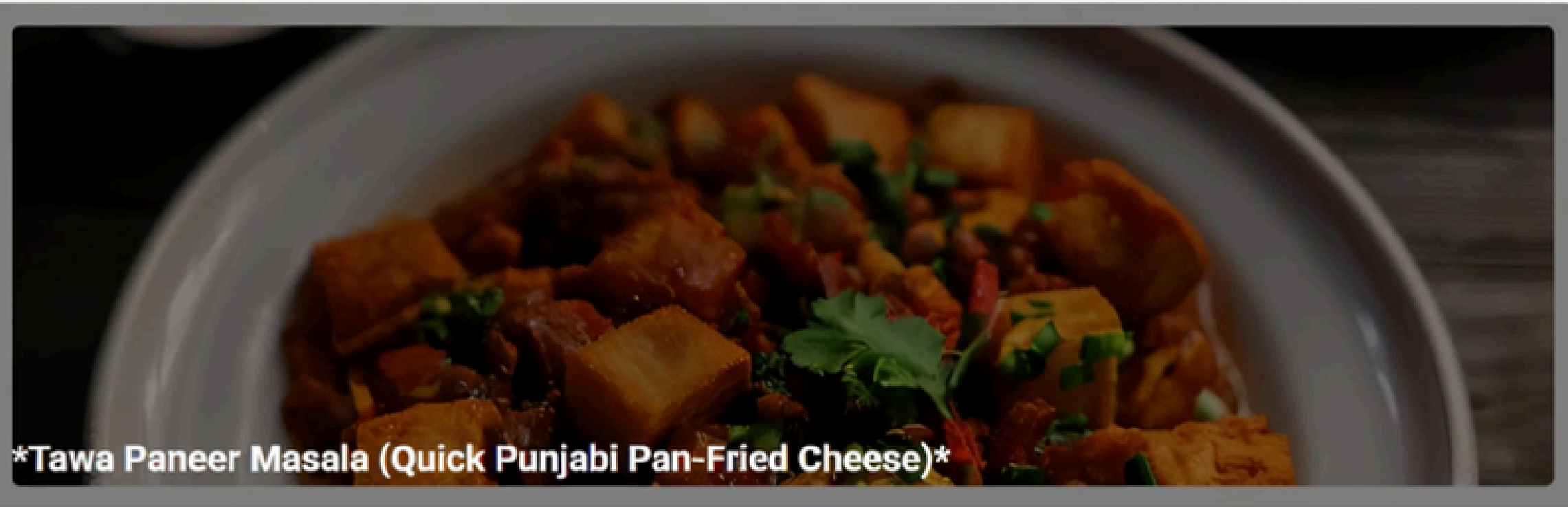
More

Unwanted Ingredients

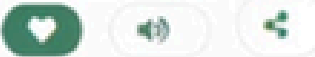
Comma-separated

Required Ingredients

Comma-separated



Tawa Paneer Masala (Quick Punjabi Pan-Fried Cheese)



 21:50 mins

 Serves 2

 1565 kcal

Ingredients

Instructions

Tips

Nutrients

Add to Shopping List	Ingredients	Quantity	Calories
☐	Paneer (Indian Cheese)	250g	approx. 860 calories
☐	Onion	1 medium	approx. 44 calories
☐	Tomato	2 medium	approx. 44 calories
☐	Green Bell Pepper (Capsicum)	½ medium	approx. 15 calories
☐	GingerGarlic Paste	1 tablespoon	approx. 15 calories
☐	Green Chilies	12	approx. 1 calorie
☐	Turmeric Powder	½ teaspoon	approx. 3 calories
☐	Red Chili Powder	1 teaspoon	approx. 6



Calories

237

12% of daily goal



Protein

4g

9% of daily goal



Carbs

11g

4% of daily goal



Fat

5g

7% of daily goal



Add Food

milk



Milk Bread Buns

Calories: 185

Protein: 8g

Carbs: 31g

Fat: 3g

Breakfast ▾

1

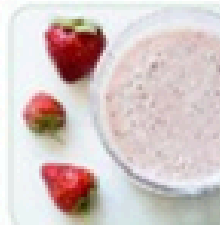


Vitamin A: 67 IU

Vitamin C: 0 mg

Calcium: 152 mg

Iron: 2 mg



Strawberry Milk

Calories: 189

Protein: 2g

Carbs: 40g

Fat: 4g

Breakfast ▾

1

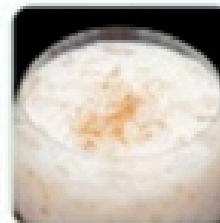


Vitamin A: 47 IU

Vitamin C: 13 mg

Calcium: 715 mg

Iron: 2 mg



Milk Punch Recipe

Calories: 237

Protein: 4g

Carbs: 11g

Fat: 5g

Breakfast ▾

1



Vitamin A: 65 IU

Vitamin C: 0 mg

Calcium: 160 mg

Iron: 0 mg



Milk Punch

Calories: 243

Protein: 1g

Carbs: 17g

Fat: 1g

Breakfast ▾

1



Vitamin A: 16 IU

Vitamin C: 0 mg

Calcium: 43 mg

Iron: 1 mg



Your Weekly Meal Plan

Personalized nutrition to help you reach your health goals

SUN

16

MON

17

TUE

18

WED

19

THU

20

FRI

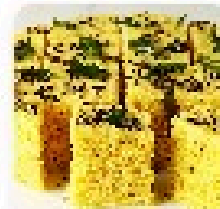
21

SAT

22

Sunday, 16 March

BREAKFAST



Dhokla

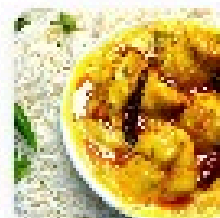
Calories: 150 kcal

Carbs: 25g

Protein: 6g

Fats: 3g

LUNCH



Kadhi Chawal

Calories: 400 kcal

Carbs: 60g

Protein: 15g

Fats: 10g

DINNER



Palak Paneer

Calories: 350 kcal

Carbs: 20g

Protein: 20g

Fats: 15g





Tailored Nutrition for Better Health

Find diet plans to help manage your health conditions



Find Disease by Its First Letter

A

B

C

D

E

F

G

H

I

J

K

L

M

N

O

P

Q

R

S

T

U

V

W

X

Y

Z



Dengue

Overview

Dengue is a viral infection caused by mosquitoes, causing flu-like symptoms like joint pain, headache, high fever, skin rashes, nausea, and vomiting. The disease is rapidly emerging and affects urban poor areas, suburbs, and rural areas. The virus has four serotypes, and the infected person serves as a carrier for uninfected mosquitoes. The virus circulates in the person's blood for 2-7 days, leading to fever and increasing severity if not treated properly. Recovering from the infection can lead to lifetime immunity to one serotype, but partial immunity for other serotypes.

[Symptoms & Causes](#)[Precautions](#)[Sample Meal Plan](#)[Diet Foods](#)

♥ Symptoms

- High fever and severe headache
- Joint and muscle pain (often called "breakbone fever")
- Rash and skin redness
- Fatigue and nausea
- Bleeding gums or nose in severe cases

⚙ Causes

- Infection by the dengue virus transmitted through mosquito bites (*Aedes aegypti*)

⚠ As doctor consultation is important, this is a temporary suggestion and should not replace professional medical advice.

Welcome to NutriPurna!

I'm your AI nutritionist. Ask me about healthy foods, meal plans, dietary advice, or anything related to nutrition and wellness.

You can type your question or use the microphone to speak.

Suggest a healthy breakfast

Weight loss tips

High-protein meals

Vitamins for immunity

Namaste! I'm NutriVoice, your AI nutrition assistant. I can help you with meal plans, dietary advice, healthy recipes, and all things nutrition. How can I assist you today?

08:09 pm

Read Aloud Copy

Weight loss tips

08:09 pm

1. Eat a balanced and healthy diet (avoid processed foods) 2. Exercise regularly (at least 30 minutes daily) 3. Get enough sleep (7-8 hours per night) 4. Stay hydrated (drink 8-10 glasses of water daily) Remember, sustainable weight loss is about lifestyle changes, not quick fixes.

08:09 pm

Read Aloud Copy

Type your nutrition question here...



IN Healthier Indian Food Alternatives

Discover nutritious twists on your favorite foods

pizza



Get Alternatives

Try: biryani, burger, fries, ice cream, etc.

Suggestions

Instead of **pizza**, try these healthier Indian versions:



Multigrain roti pizza with paneer & veggies

Contains digestive spices



Missi roti base with mint chutney & grilled chicken

Gluten-free alternative



Cauliflower crust pizza with desi spices

Contains digestive spices



Tip: These options use whole grains, less oil, and more protein/fiber

Popular Foods to Explore:

Pizza

Burger

Biryani

Samosa

Ice Cream

Noodles

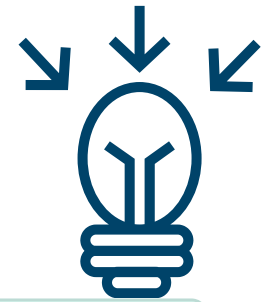
Fries

Pasta

Fried Rice

Hot Dog

Conclusion



In conclusion, through our study of existing research and analysis of similar problems in recipe generation and nutrition management, we have identified key areas for improvement and innovation. With NutriPurna, we aim to implement a comprehensive solution that combines personalized recipe discovery, advanced ingredient substitution, nutrition tracking, and meal planning tools. Our approach is based on proven methods and is designed to offer an easy-to-use experience that caters to a wide range of dietary needs and preferences. This project reflects our commitment to delivering a next-generation platform that enhances the way users manage their culinary and nutritional goals.

References

- [1] S. Chaudhary, B. Soni, A. Sindhavad, A. Mamaniya, A. Dalvi and I. Siddavatam, "Generating Indian Recipes with AI Algorithm," *2022 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*, Pune, India, 2022, pp. 1-6, doi: 10.1109/TQCEBT54229.2022.10041463.
- [2] H. Jabeen, J. Weinz and J. Lehmann, "AutoChef: Automated Generation of Cooking Recipes," *2020 IEEE Congress on Evolutionary Computation (CEC)*, Glasgow, UK, 2020, pp. 1-7, doi: 10.1109/CEC48606.2020.91856
- [3] Y. Pan, Q. Xu and Y. Li, "Food Recipe Alternation and Generation with Natural Language Processing Techniques," *2020 IEEE 36th International Conference on Data Engineering Workshops (ICDEW)*, Dallas, TX, USA, 2020, pp. 94-97, doi: 10.1109/ICDEW49219.2020.000-1.
- [4] K. Azzimani, H. Bihri, A. Dahmi, S. Azzouzi and M. E. H. Charaf, "An AI Based Approach for Personalized Nutrition and Food Menu Planning," *2022 IEEE 3rd International Conference on Electronics, Control, Optimization and Computer Science (ICECOCS)*, Fez, Morocco, 2022, pp. 1-5, doi:10.1109/ICECOCS55148.2022.9983099.
- [5] Sharma, Tript, Utkarsh Upadhyay, and Ganesh Bagler. "Classification of Cuisines from Sequentially Structured Recipes." *2020 IEEE 36th International Conference on Data Engineering Workshops (ICDEW)*, Dallas, TX, USA, 2020, pp. 72-78, doi:10.1109/ICDEW49219.2020.00008
- [6] Mao, X., Yuan, S., Xu, W. and Wei, D., 2016, December. Recipe recommendation considering the flavor of regional cuisines. In *2016 International conference on progress in informatics and computing (PIC)*, Shanghai, 2016, pp. 32-36, doi:10.1109/PIC.2016.7949462

Thank You !!